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Touhey et al.

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(54) **SELF-DEPLOYING SHELTER**

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Related U.S. Application Data

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(51) **Int. Cl.**
E04H 15/48 (2006.01)

(52) **U.S. Cl.**
CPC ***E04H 15/48*** (2013.01)

(58) **Field of Classification Search**
CPC E05H 15/48
See application file for complete search history.

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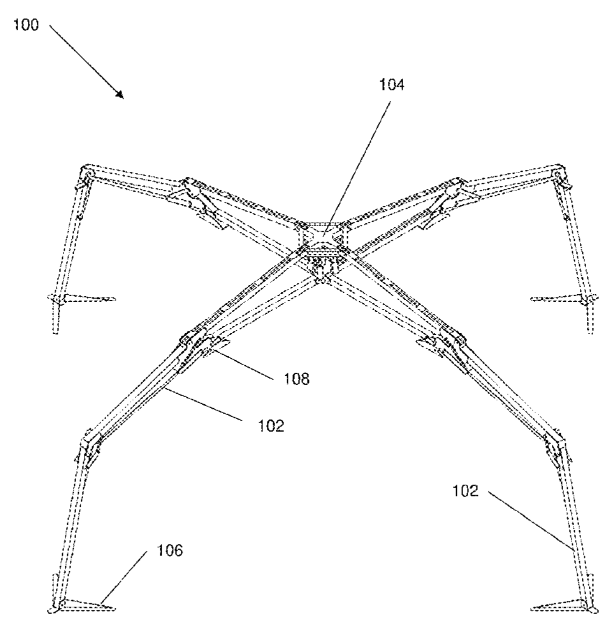
(57) **ABSTRACT**

A self-deploying shelter assembly deploys between a unfolded and folded state automatically. The shelter includes a plurality of legs and a hub, the hub being configured to be coupled to one end of a leg. A leg has a first element, a second element and a third element. A second end of the third element is movably coupled to the hub. A foot member is coupled to a first end of the first element. A second end of the first element is movably coupled to a first end of the second element and a second end of the second element is movably coupled to the first end of the third element. A roller member is coupled to one or more of the second end of the second element and the first end of the third element and is configured to contact a ground surface when the shelter is folded.

18 Claims, 28 Drawing Sheets



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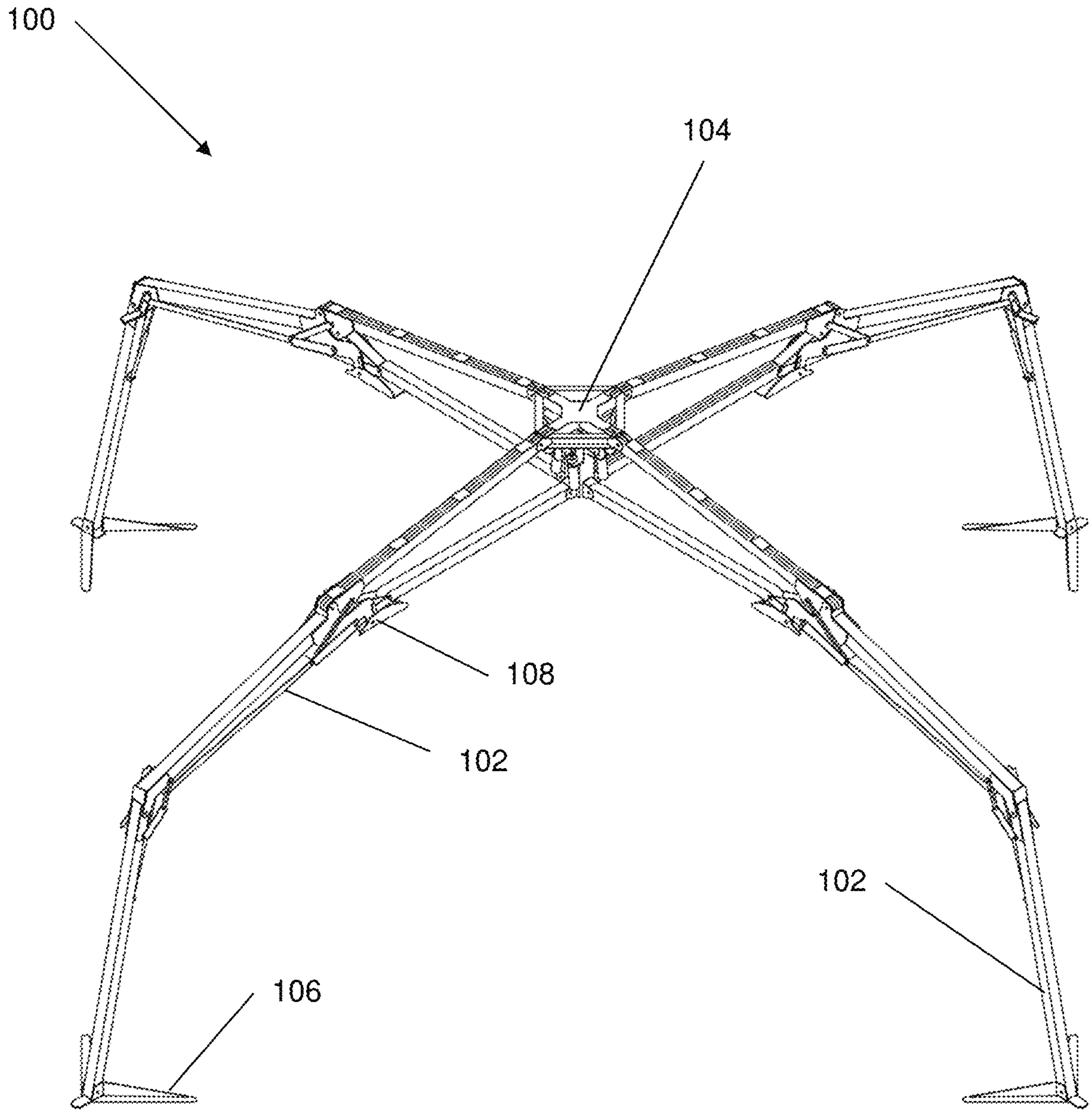

FIG. 1

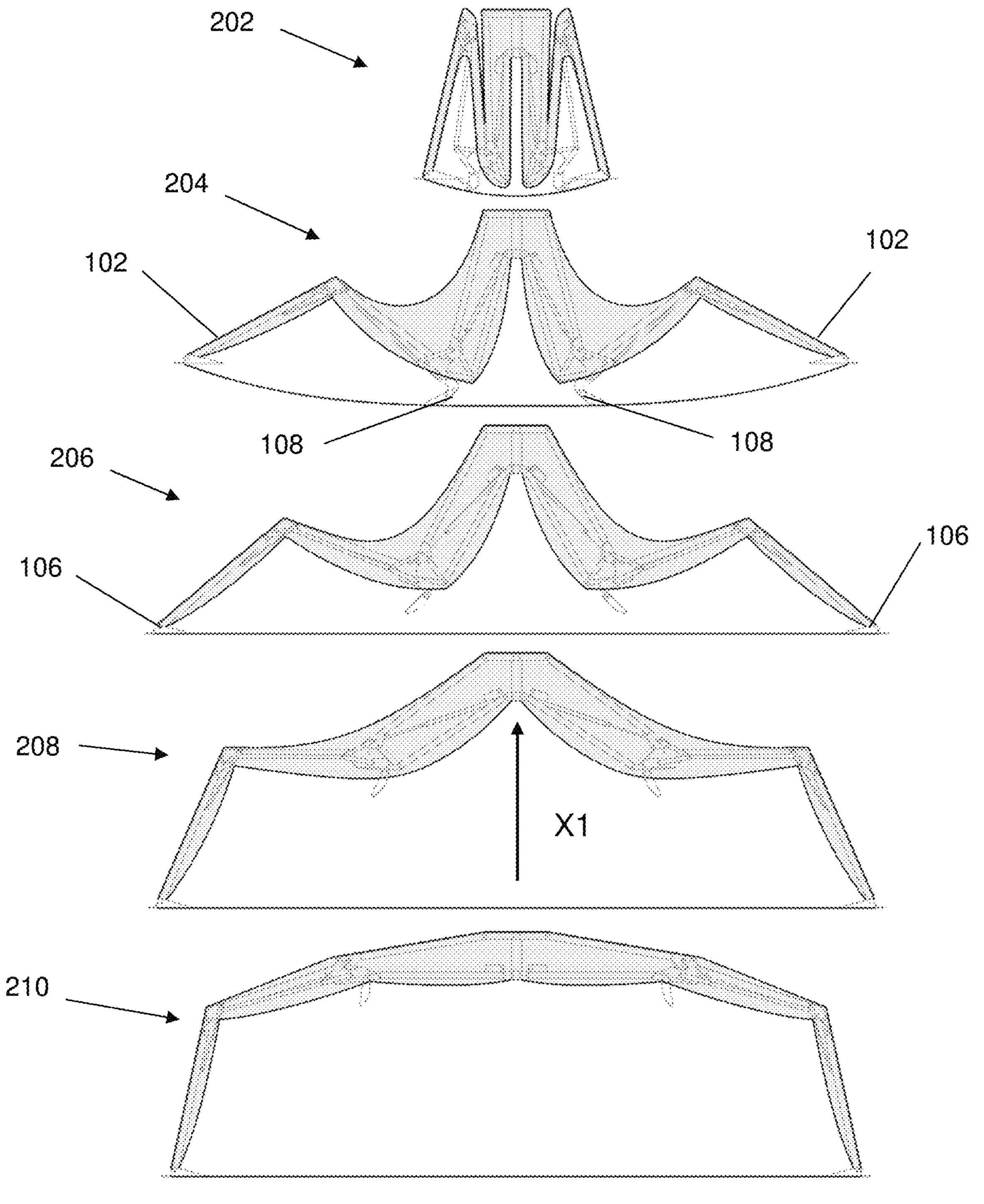

FIG. 2

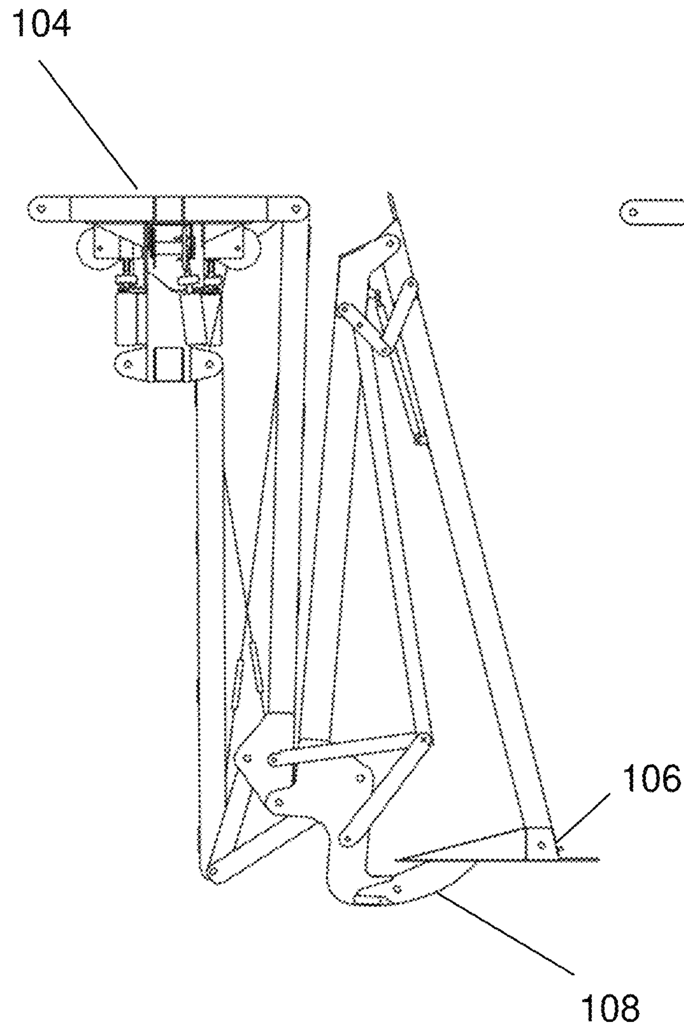

FIG. 3A

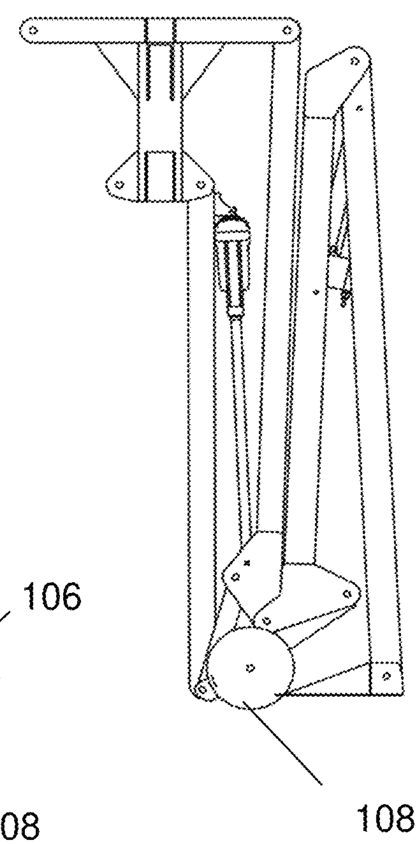

FIG. 3B

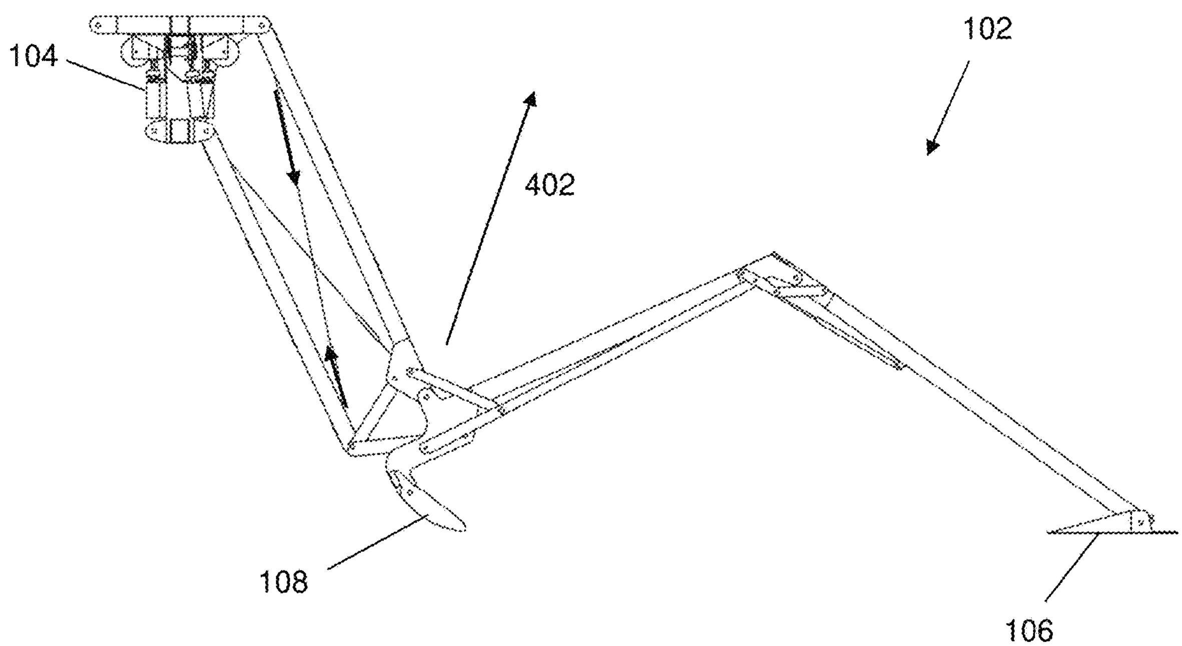

FIG. 4

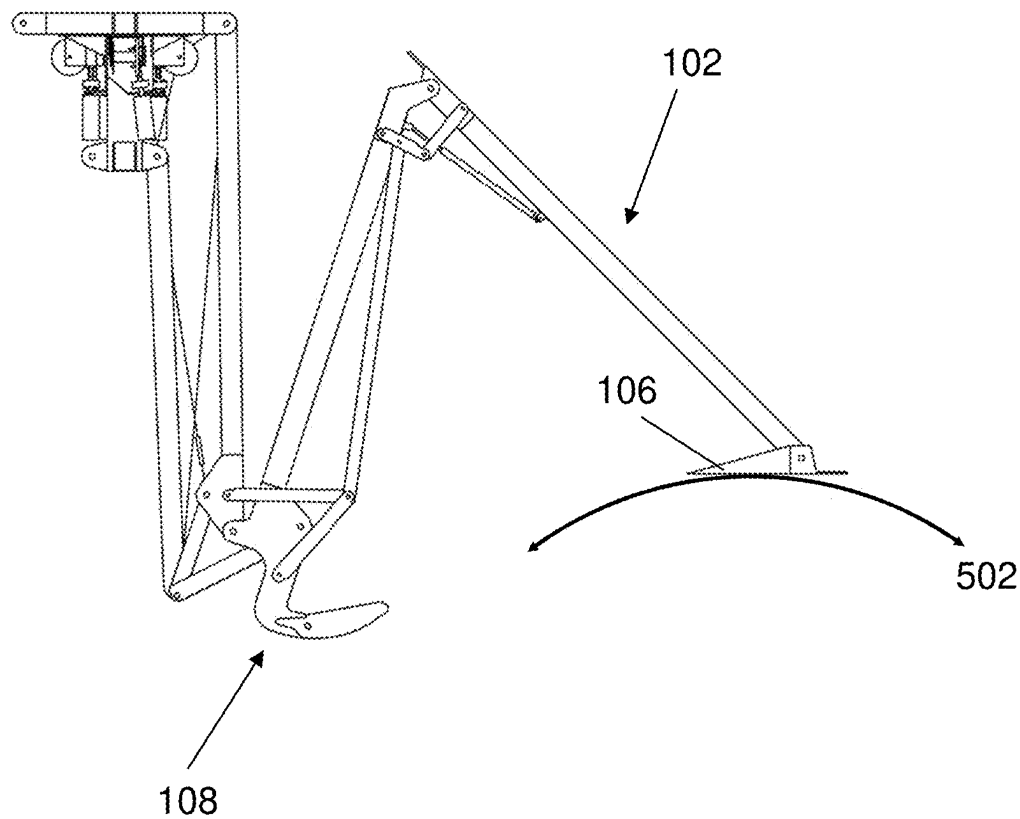

FIG. 5

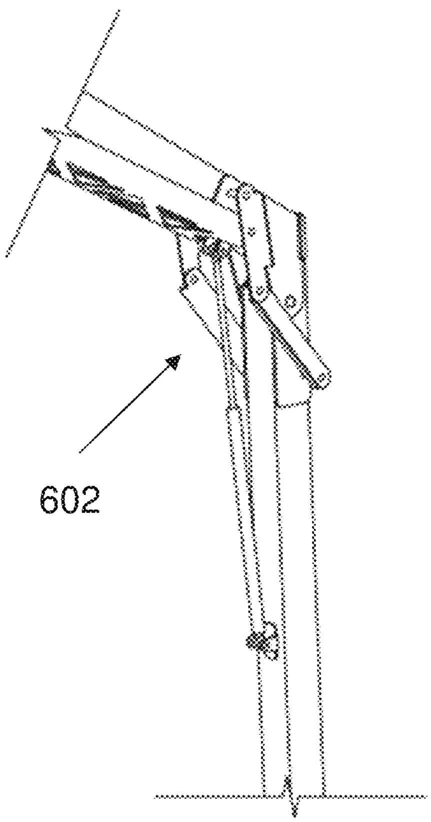

FIG. 6A

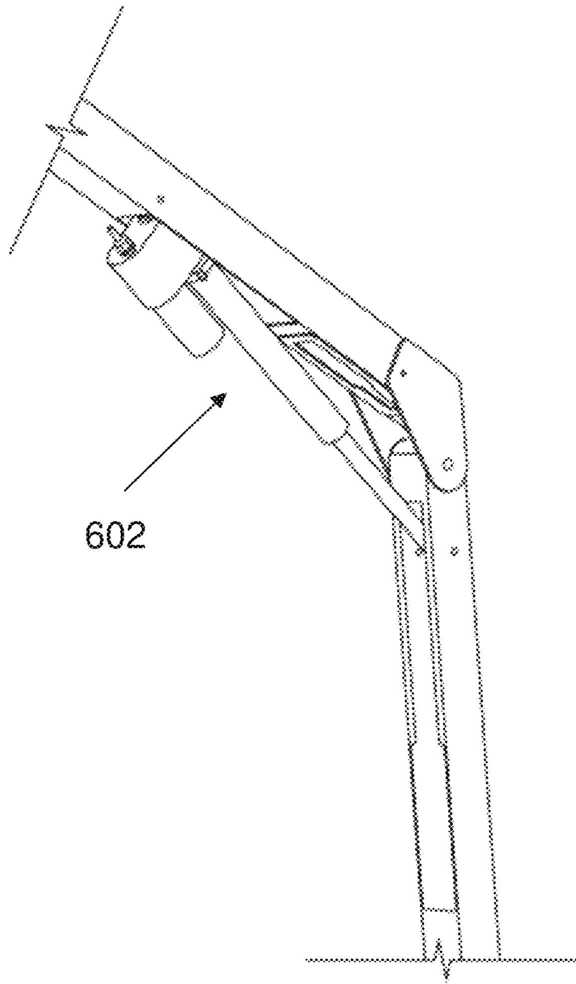

FIG. 6B

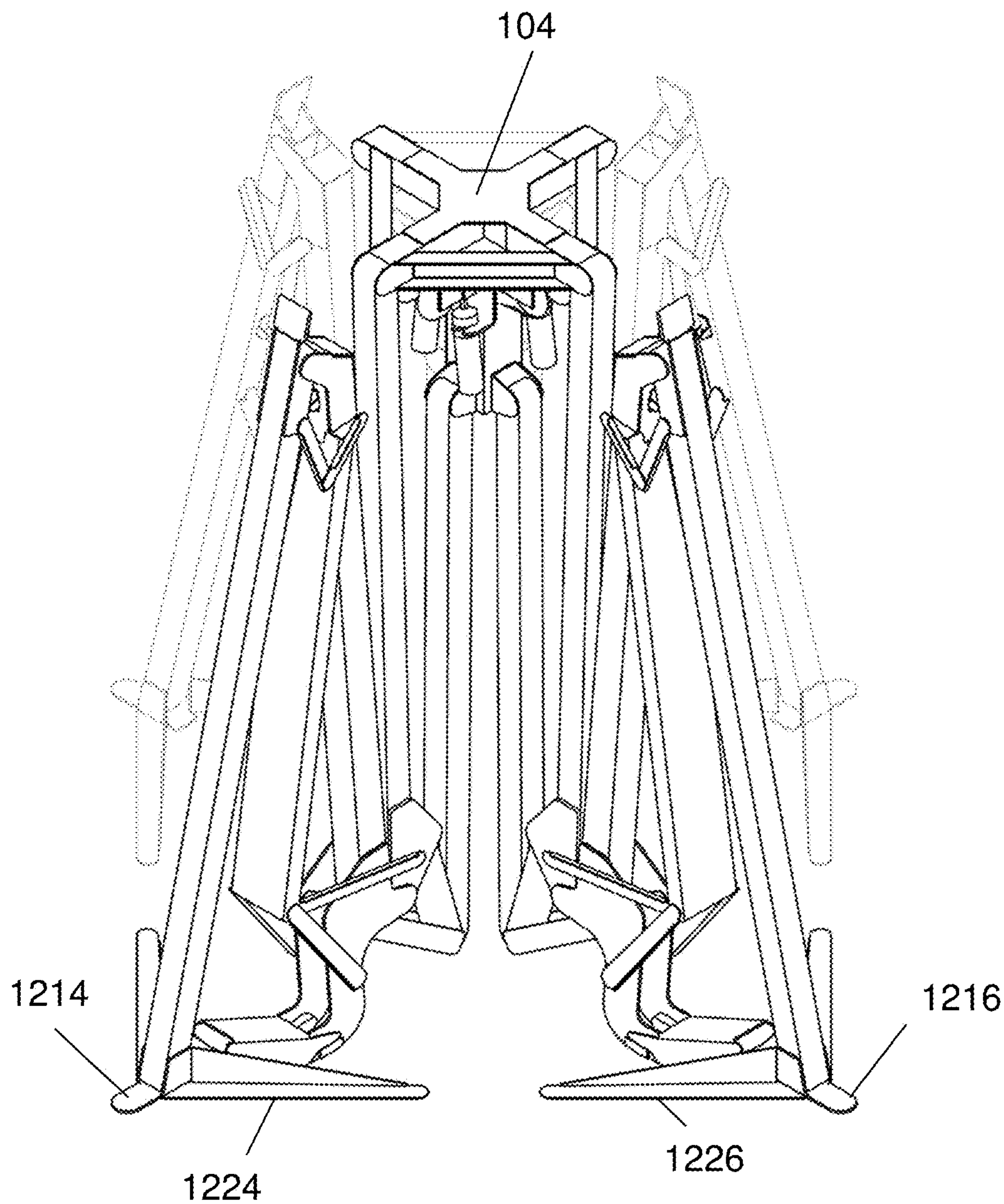

FIG. 7

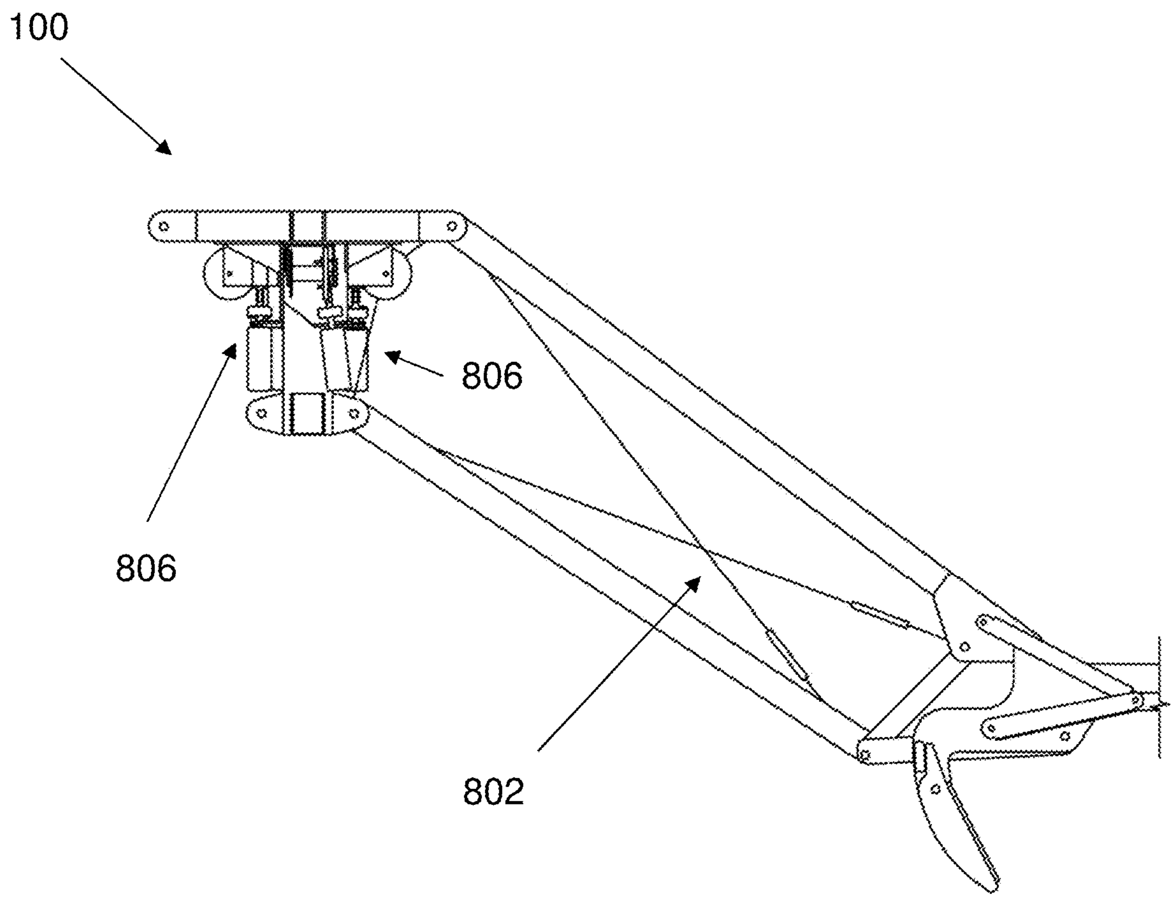

FIG. 8

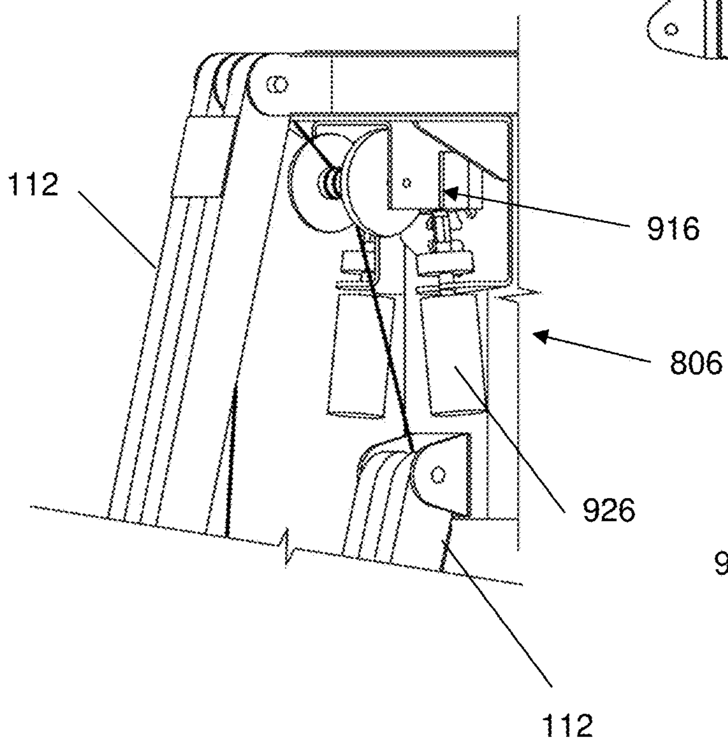

FIG. 9A

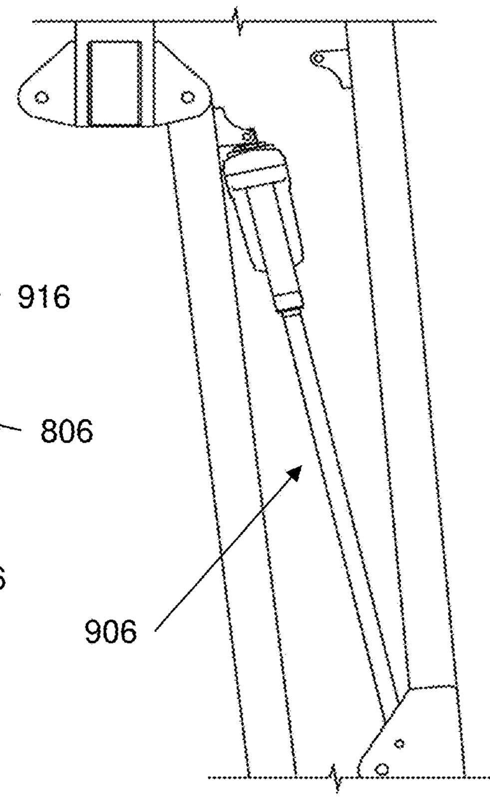

FIG. 9B

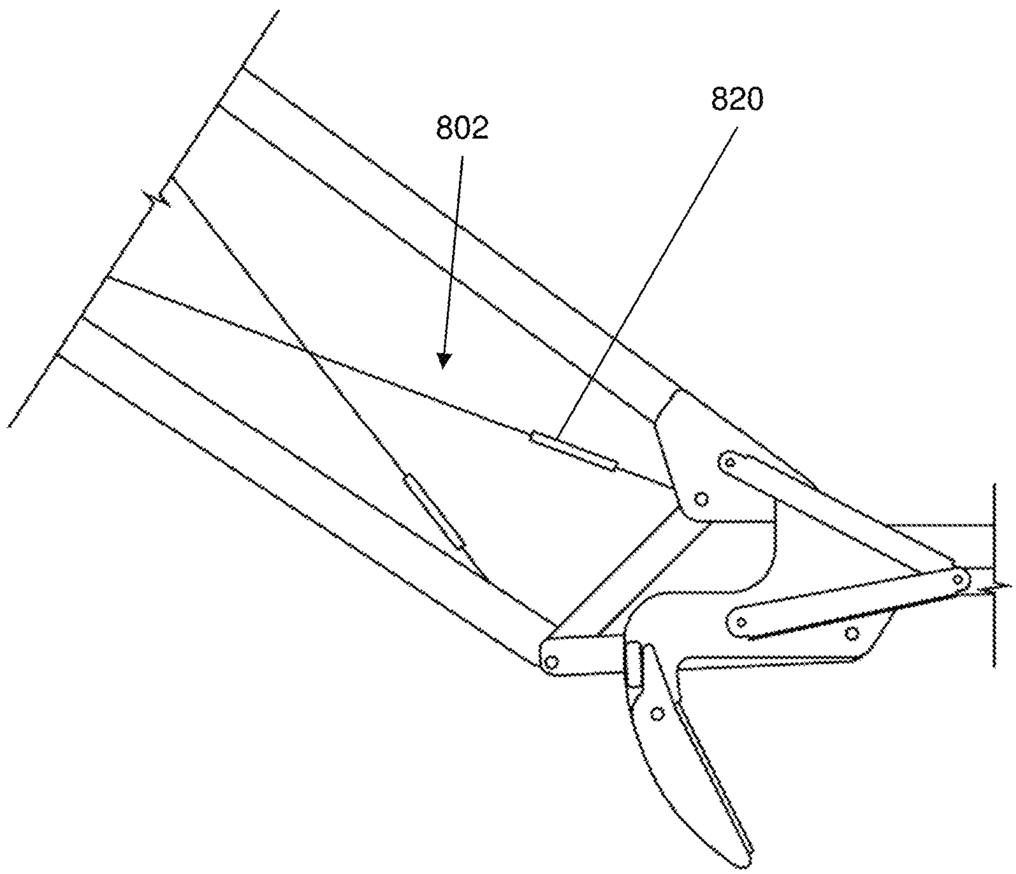

FIG. 10

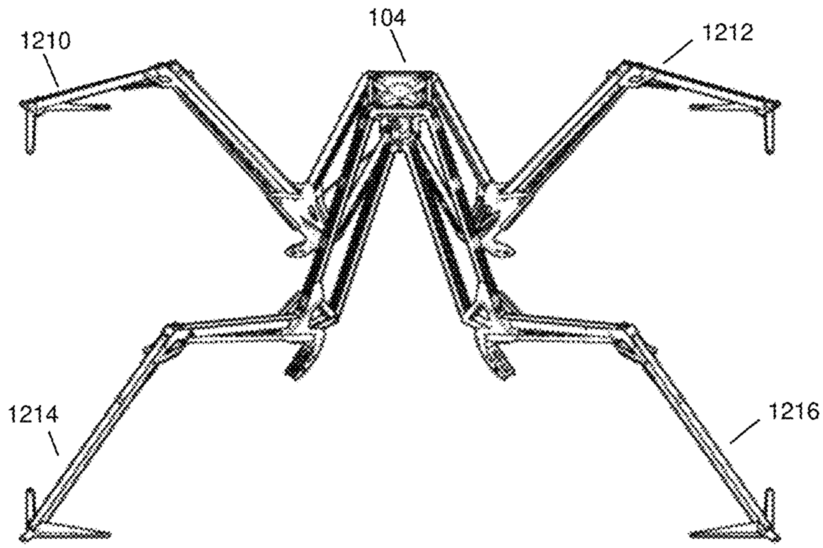

FIG. 11

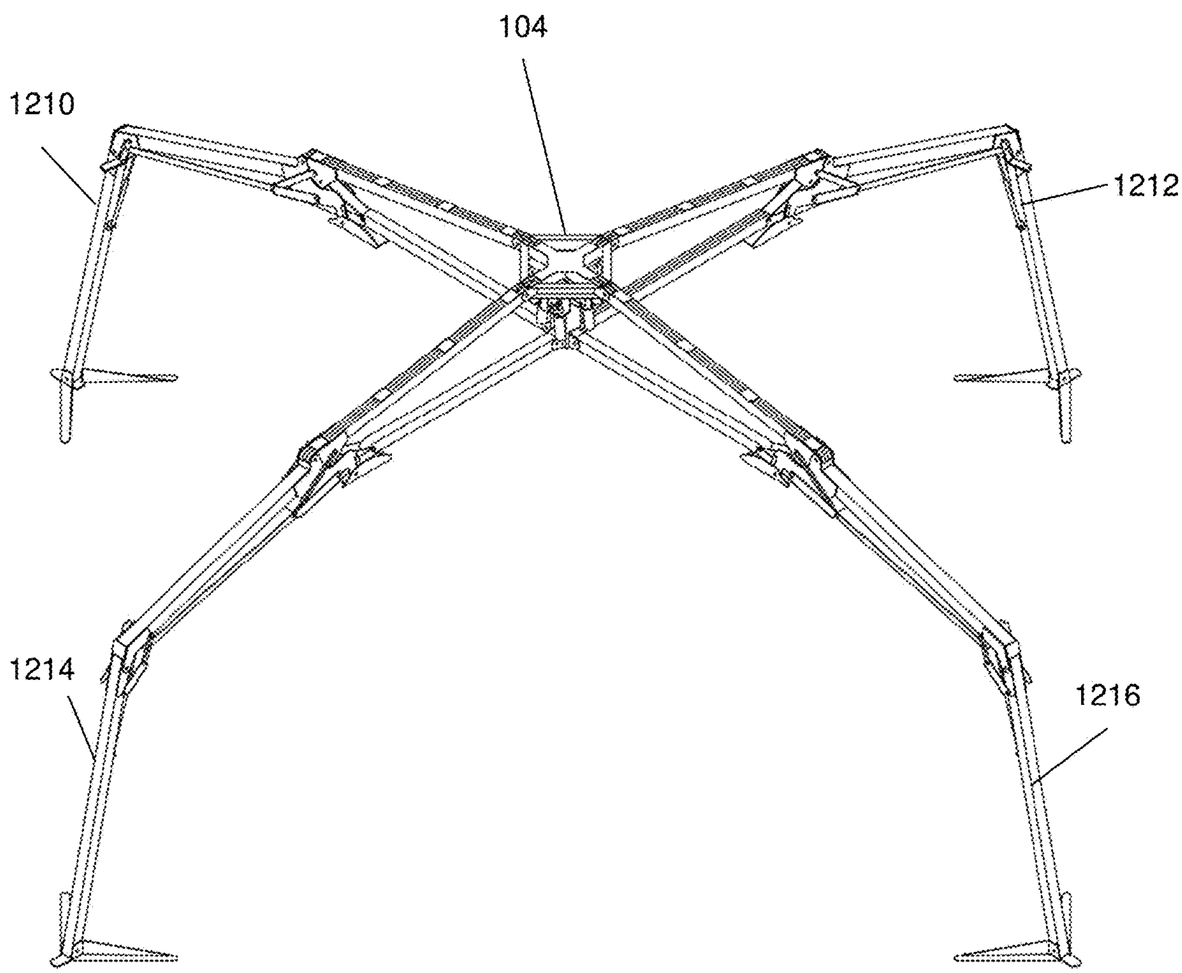

FIG. 12

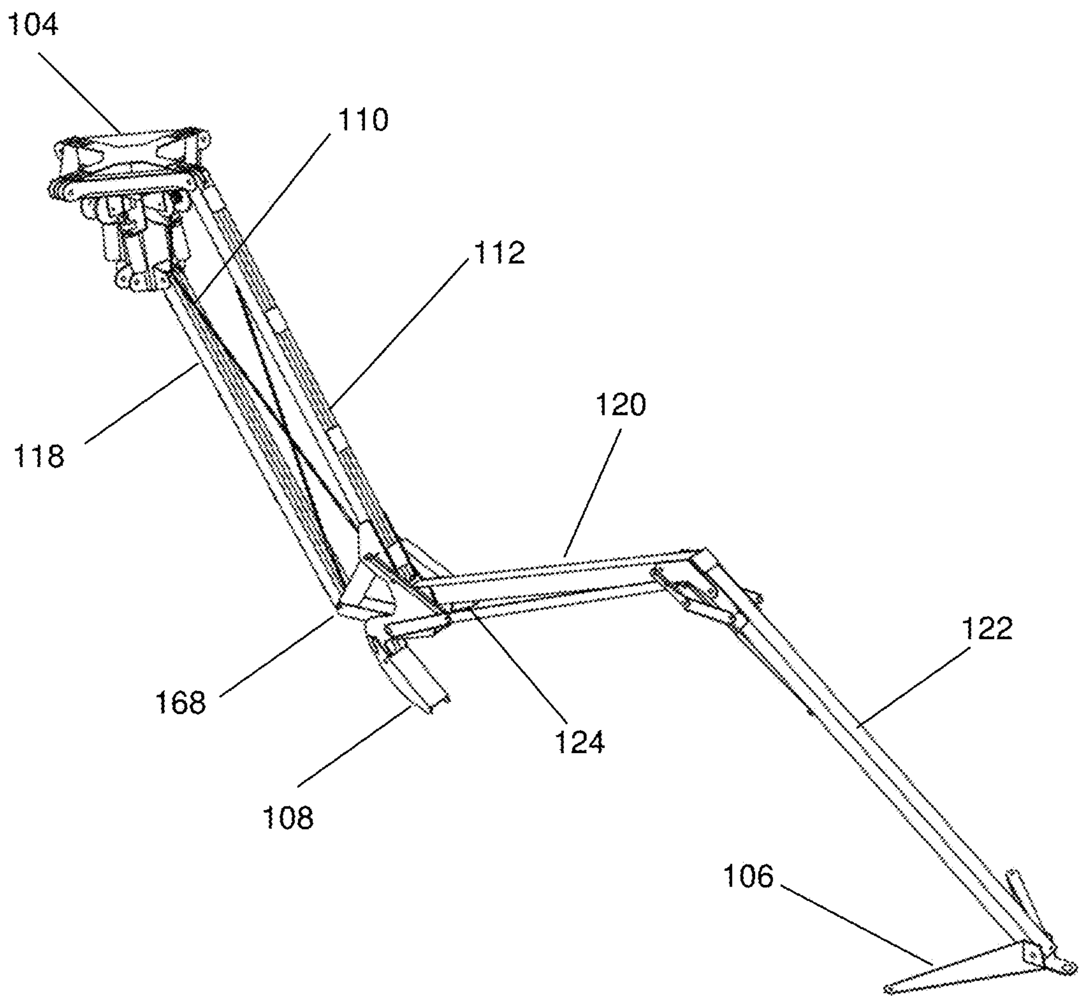

FIG. 13

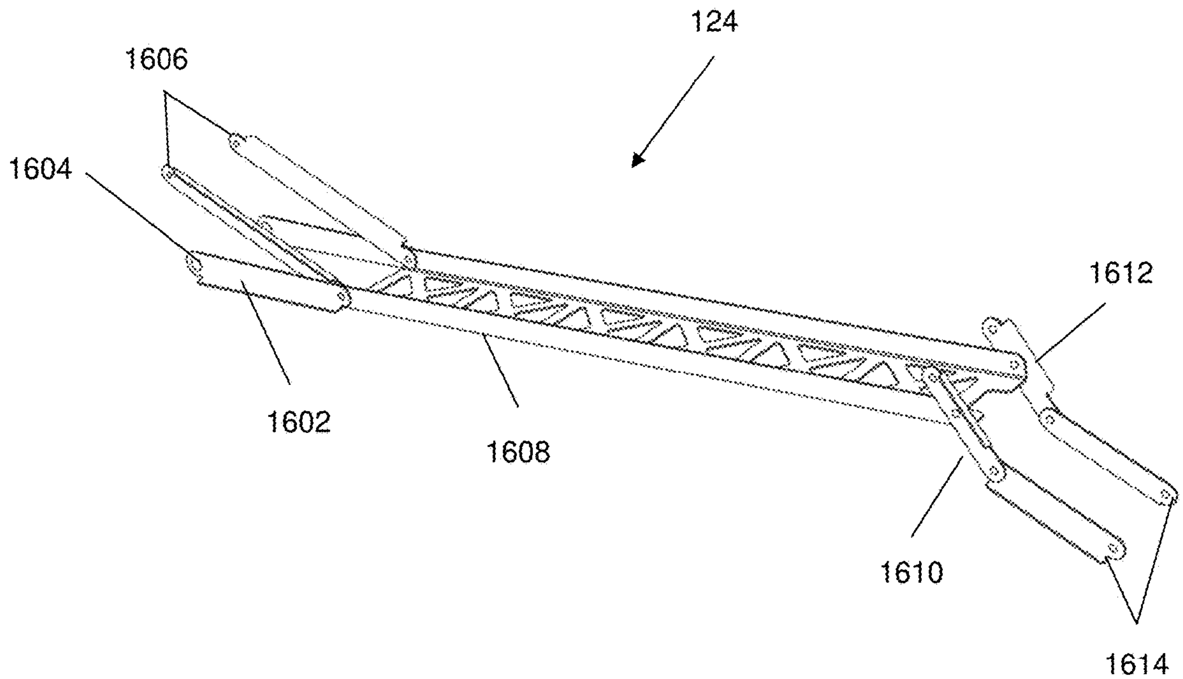

FIG. 14

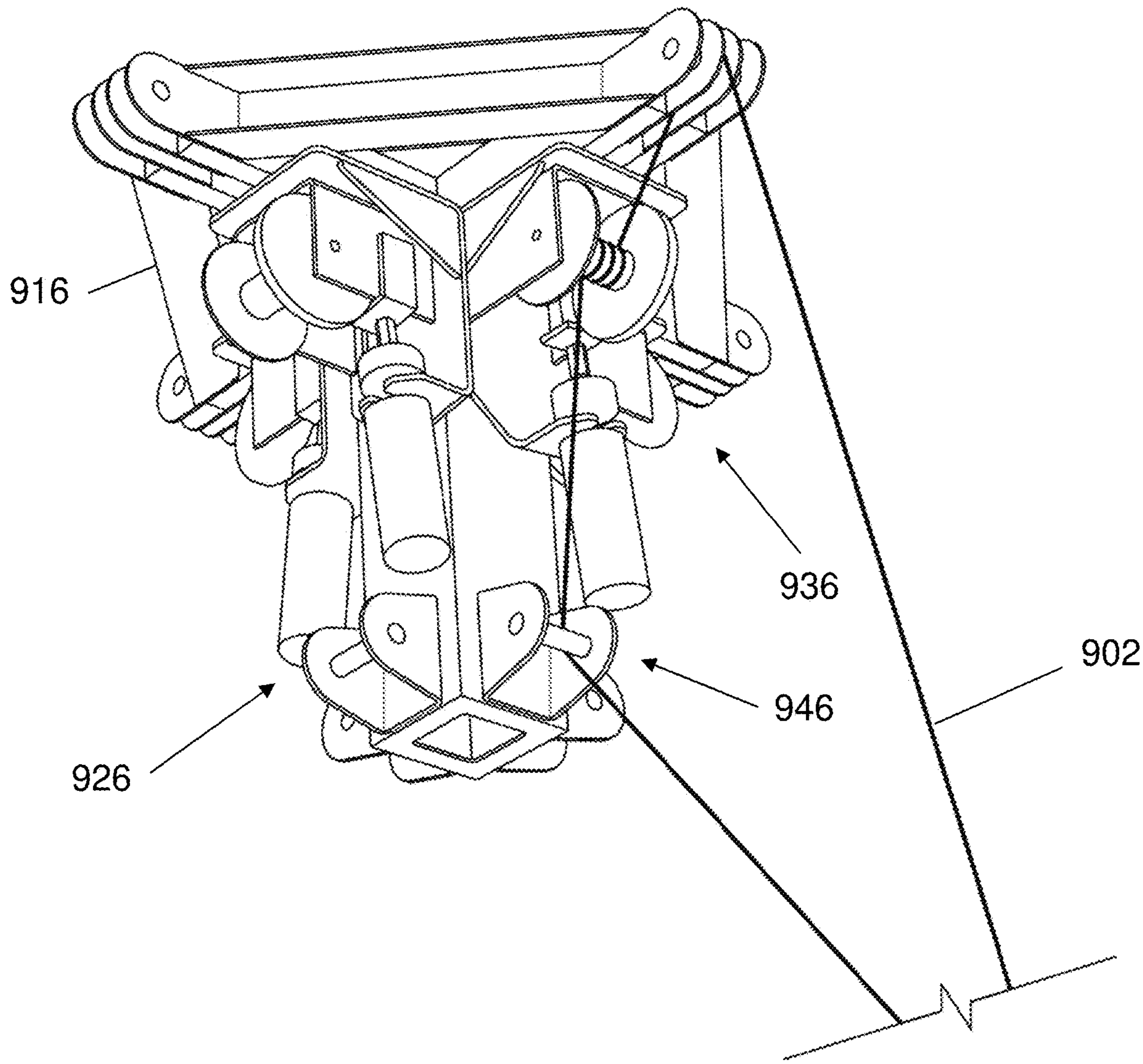

FIG. 15

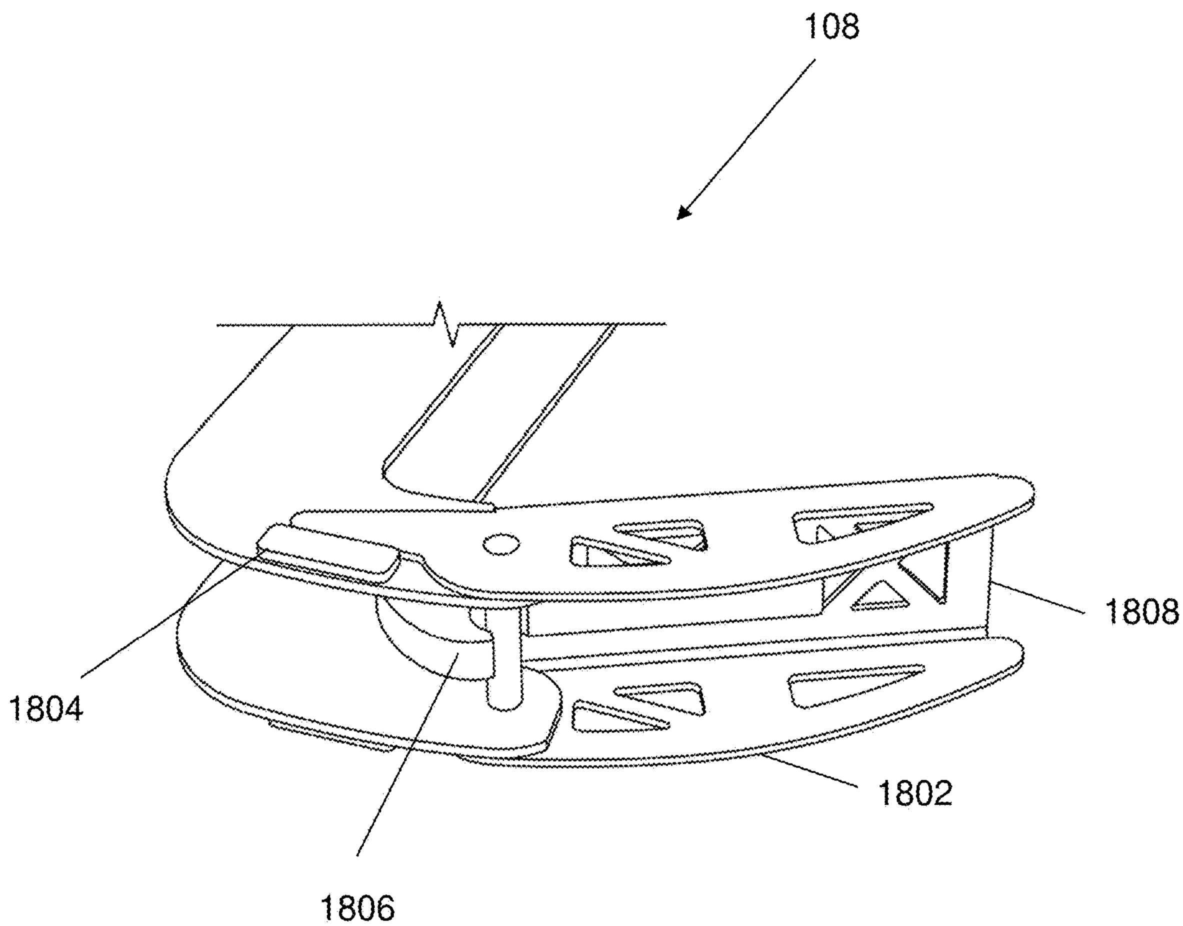

FIG. 16

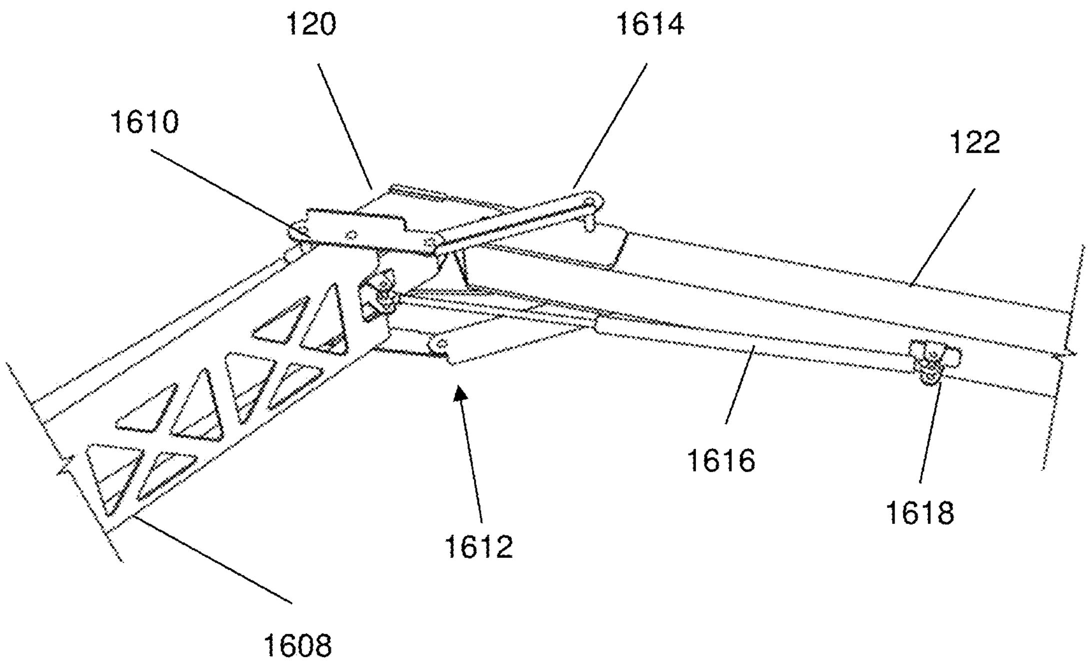

FIG. 17

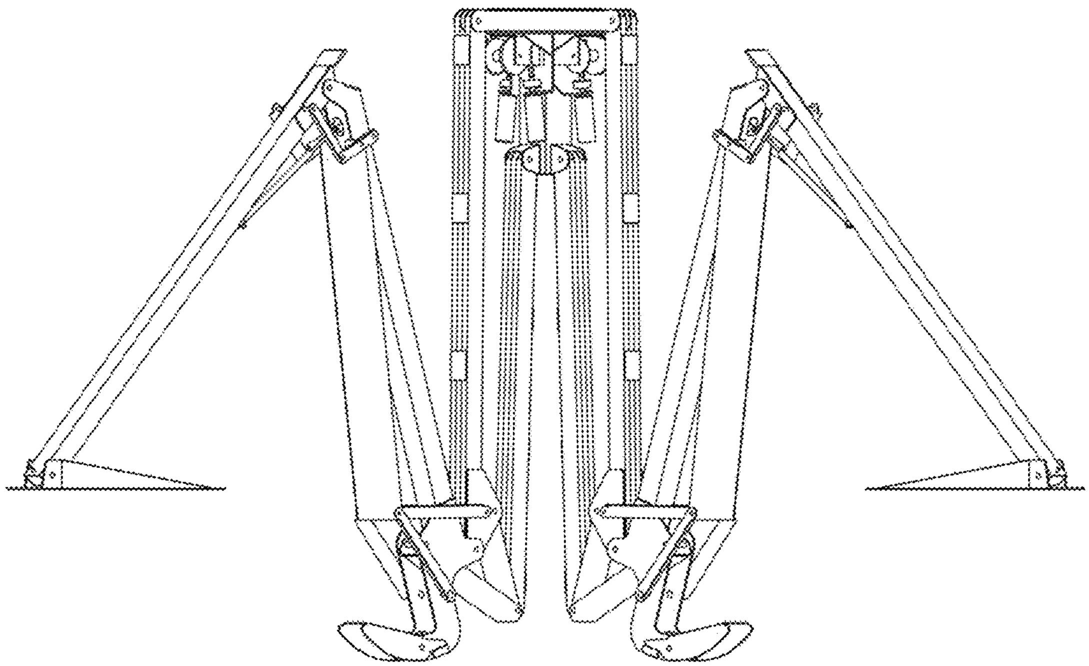

FIG. 18

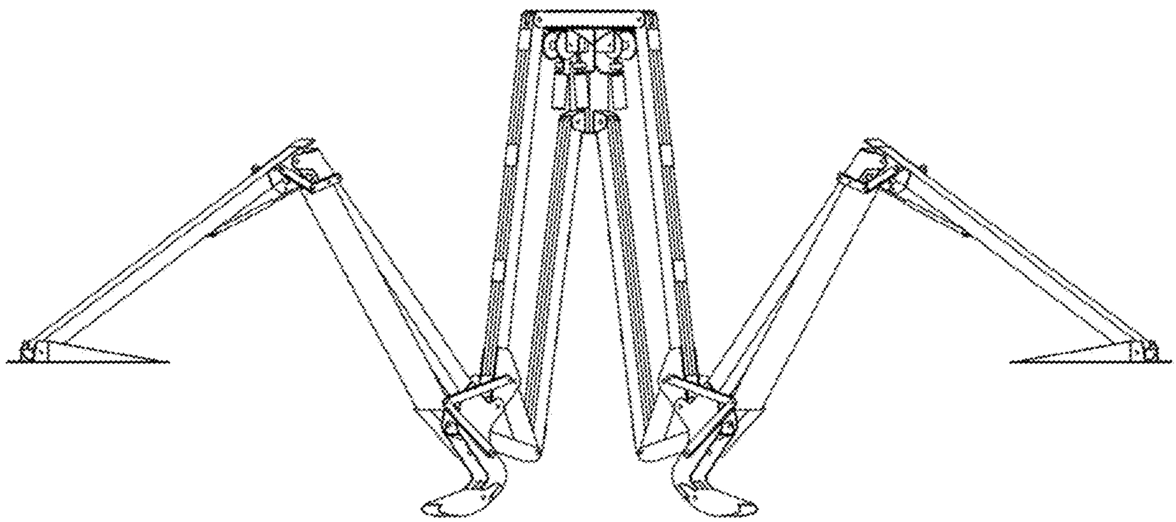

FIG. 19

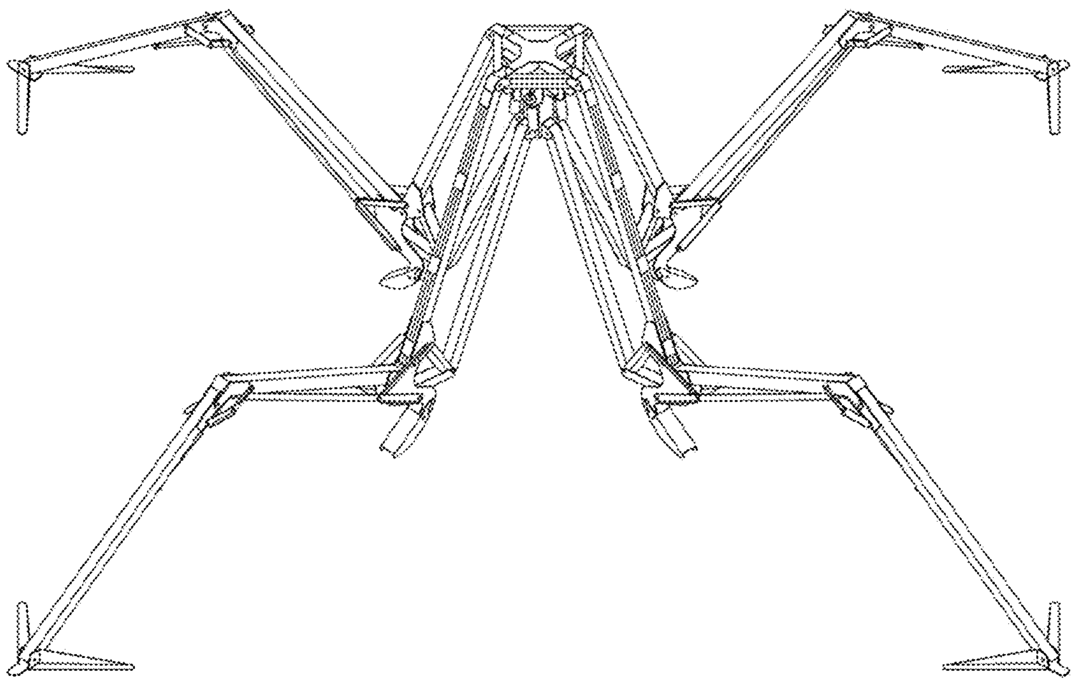

FIG. 20

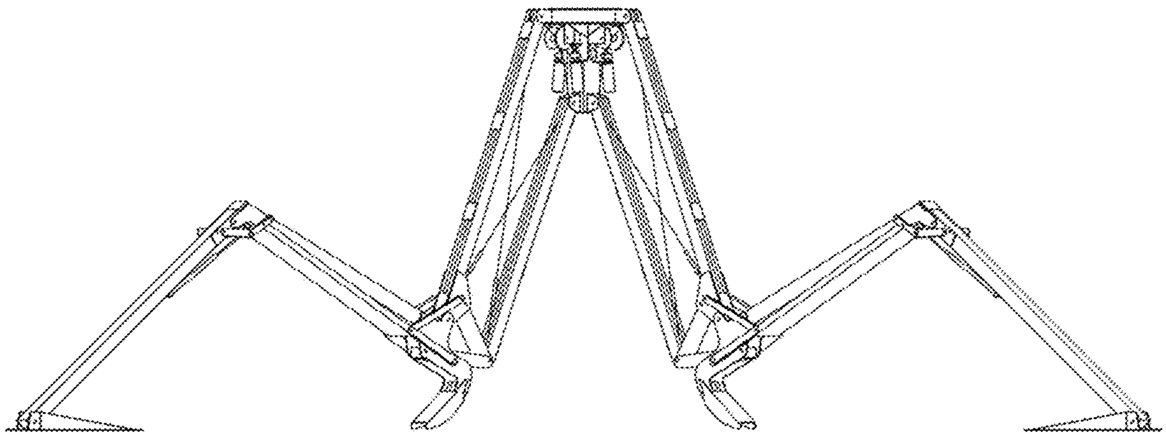

FIG. 21

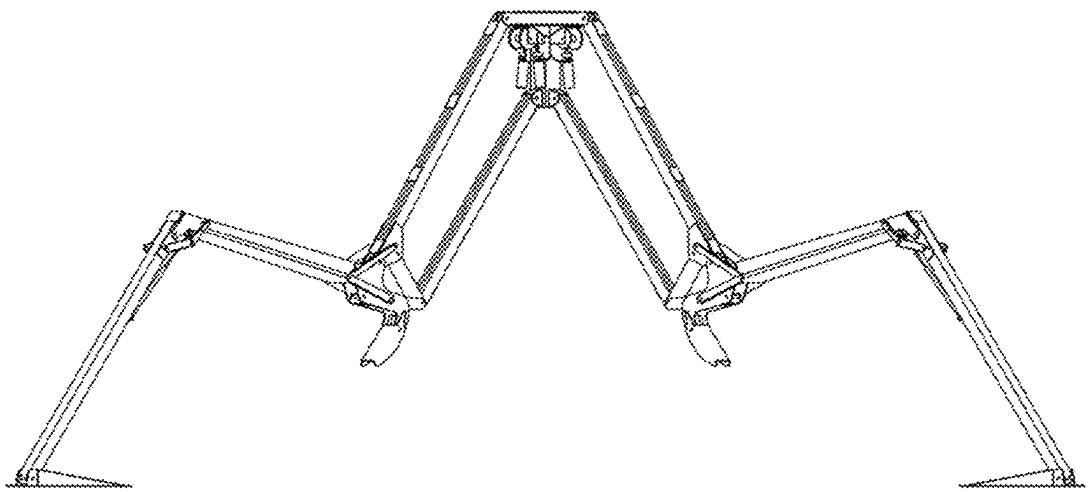

FIG. 22

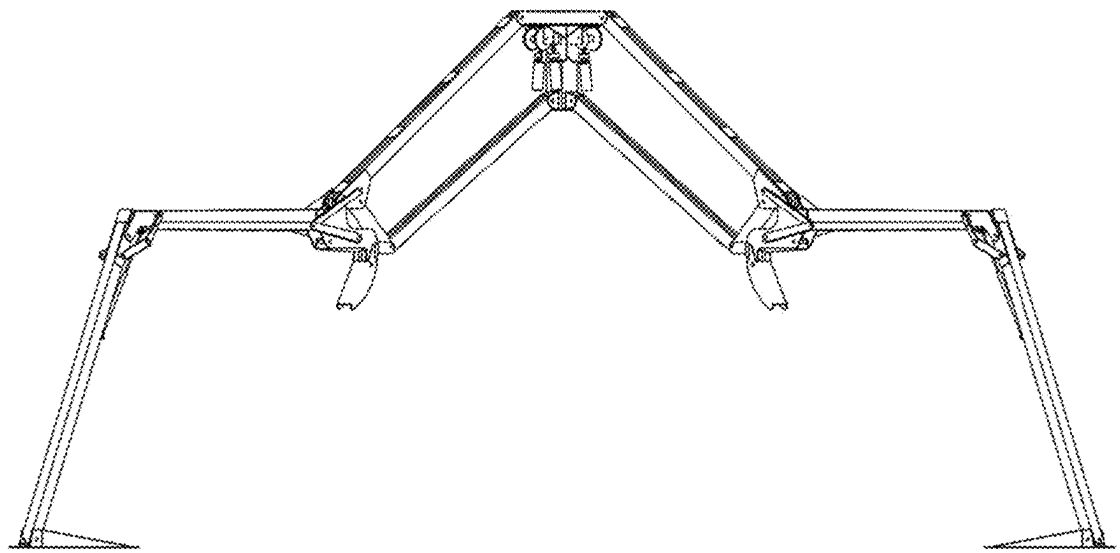

FIG. 23

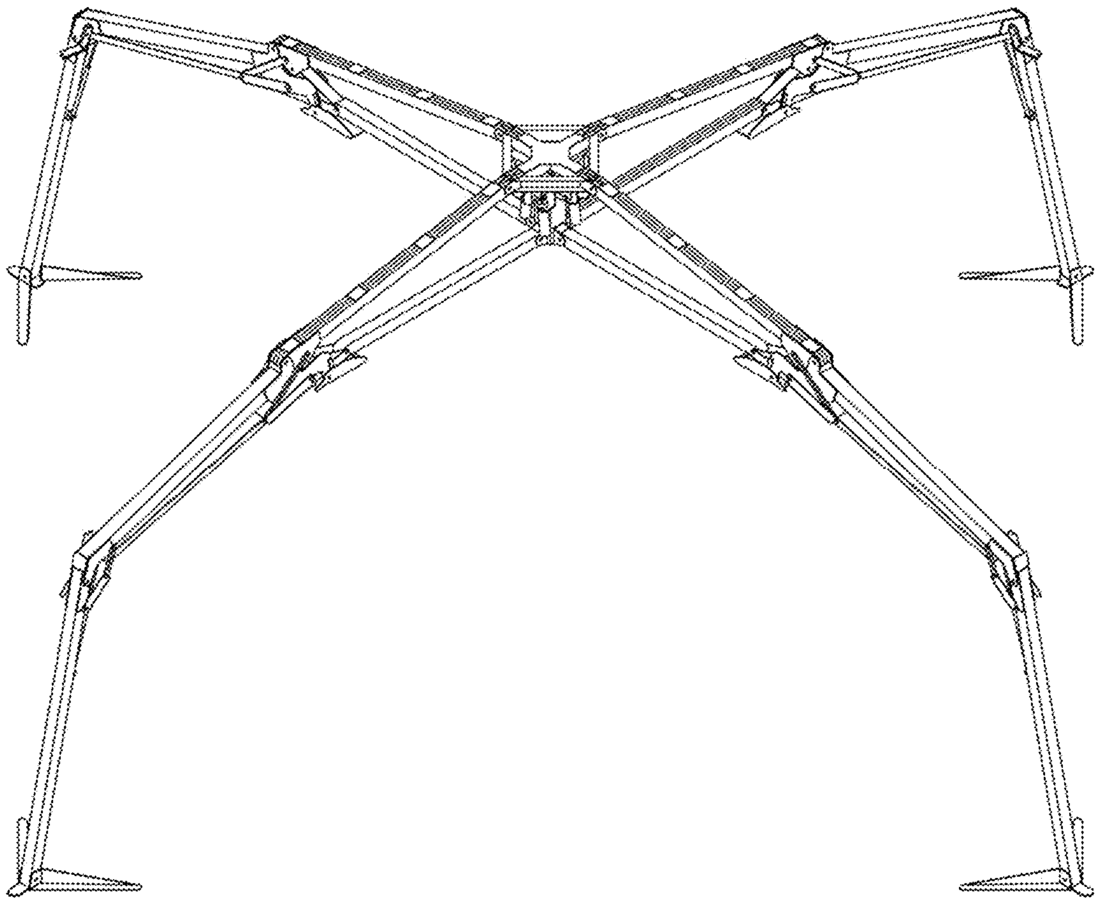

FIG. 24

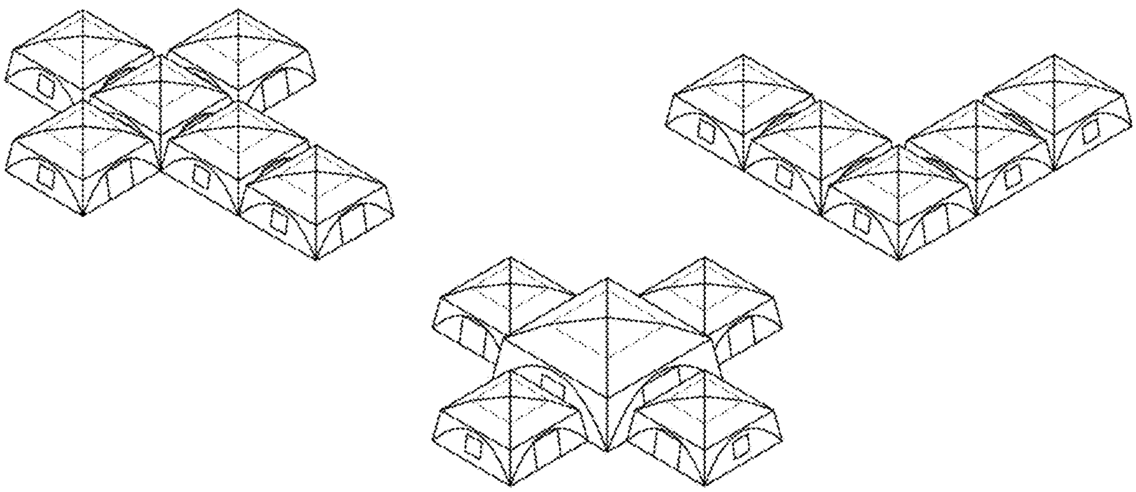

FIG. 25

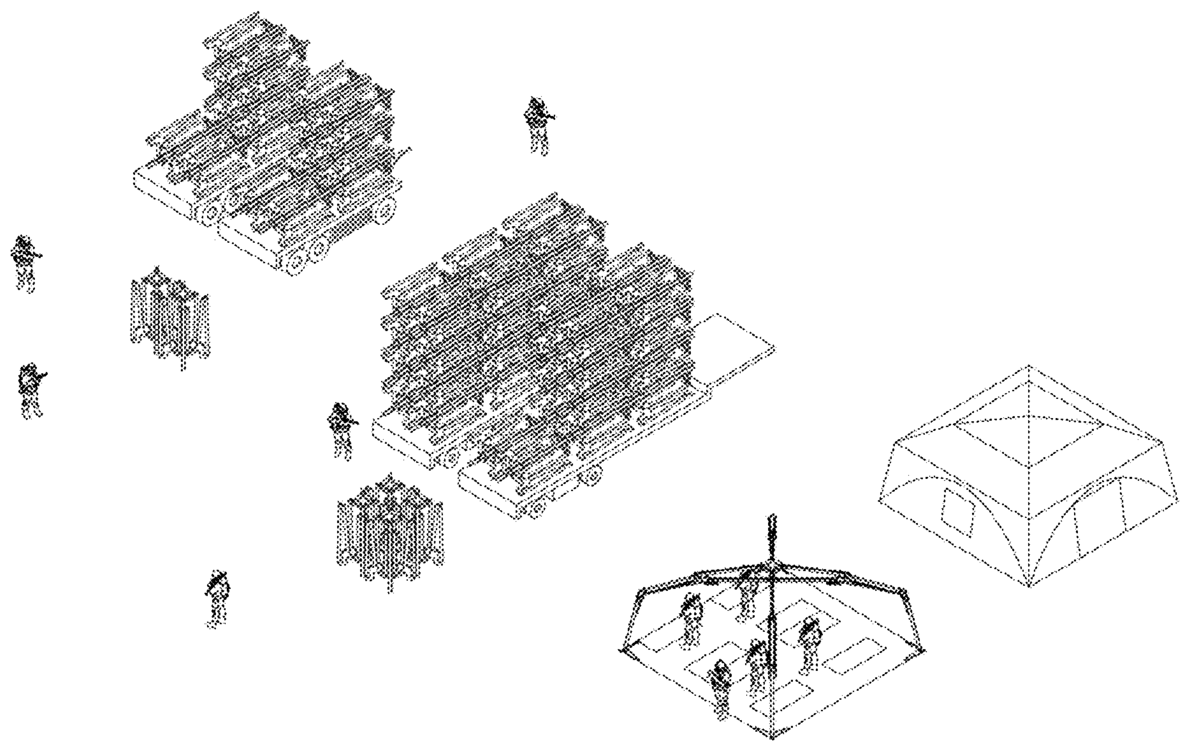

FIG. 26

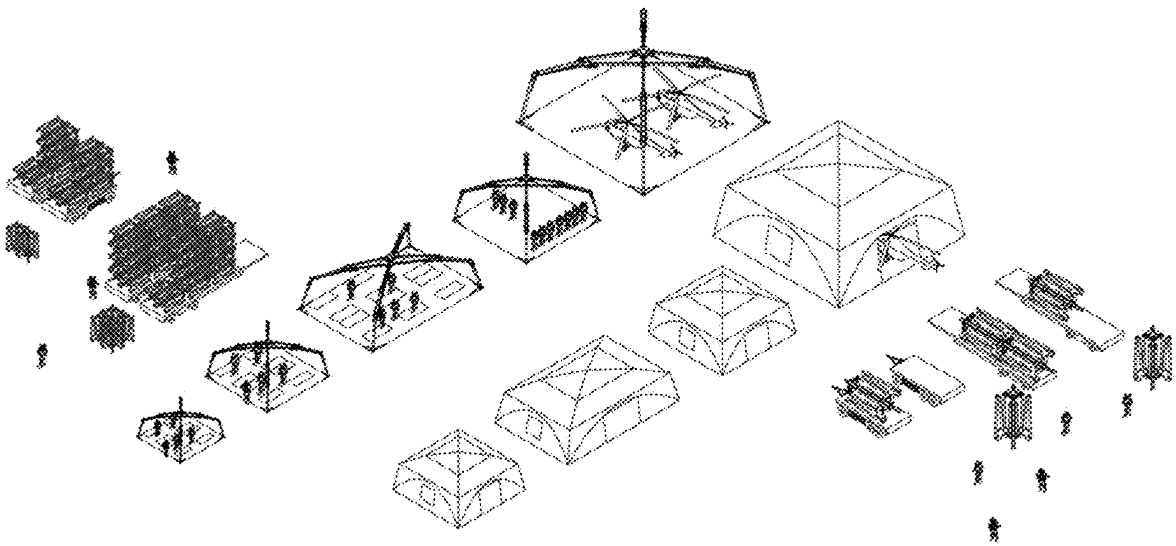

FIG. 27

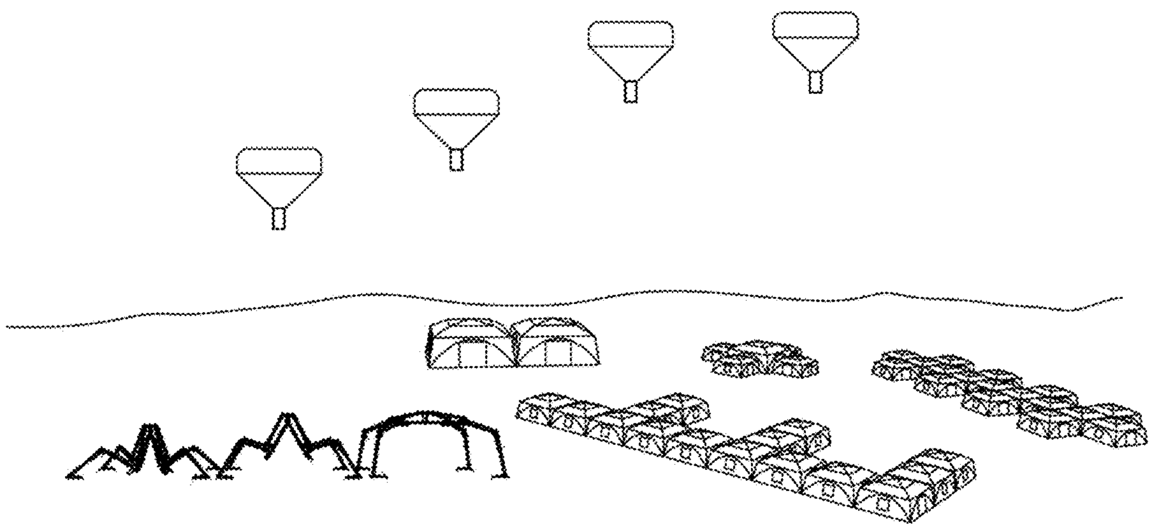

FIG. 28

SELF-DEPLOYING SHELTER

FIELD

The aspects of the disclosed embodiments generally relate to portable shelters and in particular to a fully autonomous self-deploying shelter system.

BACKGROUND

Portable shelters, also referred to as portable tents, typically require significant manpower and time to install. Historically, portable shelters are set up and torn down utilising multiple persons folding, unfold, erecting and lifting, all using manual power. These portable shelters typically include a frame and fabric. The frame has orientations that can fold, expand, telescope, or bend. The fabric is attached to the frame or can be independent. The set up generally requires knowledge of the complex setup sequence and the ability to open, erect and close or disassemble the shelter. This requires significant manpower capability.

Attempts at simplifying the process of tent installation have been made by using complex joints, scissor truss folding systems, pin joints, and inflatables. Generally, these will have a very large number of folding frame members that are subject to frequent parts damage and failure. There are typically many ground contact points that must travel across irregular ground surfaces to reach final geometry.

Thus, there is a need for improved systems, apparatus and methods for portable shelters. Accordingly, it would be desirable to a portable shelter assembly that addresses at least some of the problems described above.

SUMMARY

The aspects of the disclosed embodiments are directed to a self-deploying shelter or tent. By motorizing and remote-controlling the installation the manpower and time requirements can be significantly reduced. The Self-Deploying Shelter of the disclosed embodiments utilizes actuators in the joints and linkages that are able to be controlled by the simple push of a button. Automatic erection and takedown of the Self-Deploying Shelter of the disclosed embodiments allows the physical strength and skill level of the operator otherwise required to be reduced dramatically, saving both time and manpower. The shelter of the disclosed embodiments system does not drag its feet or legs and is configured to lift itself up with a secure base when or as the feet are in their final position, minimizing challenges on rough terrain. The shelter of the disclosed embodiments has a small number of frame members. There are no ground contact points that must travel across irregular ground. The shelter is motorized and remote-controlled.

According to a first aspect, the above and further implementations and advantages are obtained by a self-deploying shelter assembly that is configured to deploy to and from a folded state from an and to an unfolded state in an automated manner. In one embodiment, the shelter assembly includes a plurality of leg members and a hub, the hub being configured to be coupled to one end of respective leg member of the plurality of leg members. A leg member of the plurality of leg members has a first leg element, a second leg element and a third leg element. A second end of the third leg element is movably coupled to the hub. A foot member is coupled to a first end of the first leg element. A second end of the first leg element is movably coupled to a first end of the second leg element and a second end of the second leg element is

movably coupled to the first end of the third leg element. A roller member is coupled to one or more of the second end of the second leg element and the first end of the third leg element. The roller element is configured to contact a ground surface when the self-deploying shelter assembly is in the folded, non-deployed state.

In a possible implementation form, the hub is configured to cause a movement of the plurality of legs outward, away from the hub, when the shelter assembly is deploying from the folded state to the unfolded state, and cause a movement of the plurality of legs inward, toward the hub, when the shelter assembly is deploying from the unfolded state to the folded state.

In a possible implementation form, the roller member comprises a curved member that extends from a bottom of the shelter assembly to make contact with the ground surface when the shelter assembly is in the folded state.

In a possible implementation form, the foot member is away from the ground surface, or not in contact with the ground surface, in the folded state of the shelter assembly. The foot member is configured to only make contact with the ground surface once the leg members are deployed, and the hub is ready to move upward to expand the tent portion of the assembly. This way, the foot member does not drag on the ground surface and can be more accurately positioned.

In a possible implementation form, the hub member causes the plurality of leg members to extend outwards. The first leg element and the second leg element extend outwards a pre-determined distance until the foot member comes in contact with the ground surface.

In a possible implementation form, the contact of the foot member with the ground surface while the hub is extending the plurality of leg members causes the hub to move in an upwards direction, away from the ground surface. The hub moves upwards with the leg members which cause the fabric to expand over the leg members as the shelter deploys.

In a possible implementation form, the foot member is disposed closer to the hub than the roller member in the folded state of the assembly. The foot member is away from the ground surface while the roller member is in contact with the ground surface, in the folded state.

In a possible implementation form, the first leg member is pivotally connected to the foot member and the second leg member. The different elements are coupled to each other in a pivotal manner, which enables movement of the elements relative to one another.

In a possible implementation form, the second leg member is pivotally connected to the third leg member.

In a possible implementation form, the roller member comprises a bracket that is pivotally connected to a bracket on the third leg member and an arm element connected to the second leg member. The arced portion of the roller member can extend from, or be part of, the roller member.

In a possible implementation form, the foot member moves along an arced path when moving to and from the folded state and unfolded state of the shelter assembly. The movement of the foot member follows a curved path until it is moved to make contact with the ground surface.

In a possible implementation form, when the shelter assembly deploys from the folded state to the unfolded state, the hub is configured to cause the first end of the first leg member to move in an upward direction away from the ground surface, wherein the first end of the first leg member is moving along the arc path; the first end of the second leg member to move away from a centerline of the hub and third leg member; cause the foot member to come in contact with the ground surface when the first leg member and the second leg member reach pre-determined positions; cause the roller member to move away from the ground surface after the foot member comes in contact with the ground surface and the third leg member extends and cause the shelter assembly to reach the unfolded state once the third leg member is fully extended.

In a possible implementation form, when the shelter assembly deploys from the unfolded state to the folded state, the hub is configured to cause the third leg member to retract from a fully extended position until the roller member comes in contact with the ground surface; the foot member to lift off the ground surface after the roller member comes in contact with the ground surface; and cause the second leg member and the first leg member to retract to the folded state of the shelter assembly, while the foot member moves along the arc path, without further contact with the ground surface.

In a possible implementation form, the third leg member comprises a first leg element spaced apart from a second leg element, a first end of the first leg element and a first end of the second leg element pivotally connected to the hub, and a second end of the first leg element coupled to the roller arm bracket.

In a possible implementation form, an actuator or cable assembly is connected to and between the hub and the plurality of leg members. The actuator or cable assembly is configured to cause movement of one or more of the first leg element, the second leg element and the third leg element when a cable member of the cable assembly is extended and retracted by a cable spool assembly on the hub.

In a possible implementation form a drive motor is connected to the hub and the cable spool assembly, the drive motor configured to cause the cable spool assembly to extend and retract the cable member.

In a possible implementation form, a linkage system is coupled to the plurality of leg members and leg elements, the linkage system configured to cause individual leg members to open to the unfolded state and close to the folded state along a prescribed path.

In a possible implementation form, the linkage system is connected between the hub and the individual leg member of the plurality of leg members.

In a possible implementation form, the linkage system is connected to the actuator or cable assembly.

These and other aspects, implementation forms, and advantages of the exemplary embodiments will become apparent from the embodiments described herein considered in conjunction with the accompanying drawings. It is to be understood, however, that the description and drawings are designed solely for purposes of illustration and not as a definition of the limits of the disclosed invention, for which reference should be made to the appended claims. Additional aspects and advantages of the invention will be set forth in the description that follows, and in part will be obvious from the description, or may be learned by practice of the invention. Moreover, the aspects and advantages of the invention may be realized and obtained by means of the instrumentalities and combinations particularly pointed out in the appended claims.

BRIEF DESCRIPTION OF THE DRAWINGS

In the following detailed portion of the present disclosure, the aspects of the disclosed embodiments will be explained in more detail with reference to the example embodiments shown in the drawings, in which like references indicate like elements and:

FIG. **1** is a perspective view of an exemplary self-deploying shelter assembly incorporating aspects of the disclosed embodiments;

FIG. **2** illustrates the unfolding and folding of an exemplary self-deploying shelter assembly incorporating aspects of the disclosed embodiments;

FIGS. **3**A and **3**B are side views of an exemplary self-deploying shelter assembly incorporating aspects of the disclosed embodiments, with different roller members. In FIG. **3**B the roller member is a wheel.

FIG. **4** illustrates a partially unfolded or folded exemplary self-deploying shelter assembly incorporating aspects of the disclosed embodiments;

FIG. **5** illustrates an exemplary path of the foot member during folding and unfolding of an exemplary self-deploying shelter assembly incorporating aspects of the disclosed embodiments;

FIG. **6**A illustrates the use of a strut with leg elements for an exemplary self-deploying shelter assembly incorporating aspects of the disclosed embodiments;

FIG. **6**B illustrates the use of a linear actuator with leg elements for an exemplary self-deploying shelter assembly incorporating aspects of the disclosed embodiments;

FIG. **7** illustrates another example of the folded state of an exemplary self-deploying shelter assembly incorporating aspects of the disclosed embodiments;

FIG. **8** illustrates one example of a cable assembly for an exemplary self-deploying shelter assembly incorporating aspects of the disclosed embodiments;

FIG. **9**A illustrates the use of a drive motor with a cable assembly for a self-deploying shelter assembly incorporating aspects of the disclosed embodiments;

FIG. **9**B illustrates the use of a linear actuator rather than a drive motor.

FIG. **10** illustrates the use of a spring tensioner with a cable assembly for a self-deploying shelter assembly incorporating aspects of the disclosed embodiments;

FIG. **11** illustrates a partially unfolded state of the frame of a self-deploying shelter assembly incorporating aspects of the disclosed embodiments;

FIG. **12** illustrates an unfolded state of the frame of a self-deploying shelter assembly incorporating aspects of the disclosed embodiments;

FIG. **13** illustrates an example of a partially extended leg member of a self-deploying shelter assembly incorporating aspects of the disclosed embodiments;

FIG. **14** illustrates an exemplary leg member and connection elements for a self-deploying shelter assembly incorporating aspects of the disclosed embodiments;

FIG. **15** is a view of the underside of a hub for a self-deploying shelter assembly incorporating aspects of the disclosed embodiments;

FIG. **16** illustrates an exemplary roller member for a self-deploying shelter assembly incorporating aspects of the disclosed embodiments;

FIG. **17** illustrates an exemplary connection of leg elements for a self-deploying shelter assembly incorporating aspects of the disclosed embodiments;

FIGS. **18**-**25** illustrates an exemplary series of movements and states from the unfolded state to the folded state for a self-deploying shelter assembly incorporating aspects of the disclosed embodiments;

FIG. **26** illustrates an exemplary implementation of a plurality of self-deploying shelter assemblies incorporating aspects of the disclosed embodiments;

FIGS. **27-28** illustrate exemplary implementations of one or more self-deploying shelter assemblies incorporating aspects of the disclosed embodiments.

These and other aspects, implementation forms, and advantages of the exemplary embodiments will become apparent from the embodiments described herein considered in conjunction with the accompanying drawings. It is to be understood, however, that the description and drawings are designed solely for purposes of illustration and not as a definition of the limits of the disclosed invention, for which reference should be made to the appended claims. Additional aspects and advantages of the invention will be set forth in the description that follows, and in part will be obvious from the description, or may be learned by practice of the invention. Moreover, the aspects and advantages of the invention may be realized and obtained by means of the instrumentalities and combinations particularly pointed out in the appended claims.

DESCRIPTION OF THE DISCLOSED EMBODIMENTS

The aspects of the disclosed embodiments are directed to a rapidly deployable shelter assembly or system, also referred to herein as a hands-off expeditionary tent (HEXT). One example of a shelter assembly **100** incorporating aspects of the disclosed embodiments is illustrated in FIG. **1**. The shelter assembly **100** shown in FIG. **1** is configured to be a fully autonomous self-deploying shelter. One advantage of the shelter assembly **100** is that the foot members **106** travel to and from closed to open position without dragging along ground. According to the aspects of the disclosed embodiments as will be described herein, the foot members **106** lift up over the ground, and only make contact when the leg **102** is partially extended.

The self-deploying shelter assembly **100** of the disclosed embodiments is configured to automatically, without manual intervention, deploy to and from a folded state from an and to an unfolded state. In one embodiment, the shelter assembly **100** includes a plurality of respective leg members **102** and a hub **104**. The hub **104** is configured to be coupled to one end of a respective leg member **102** of the plurality of leg members.

A leg member **102** of the plurality of leg members has a first leg element, a second leg element and a third leg element. A second end of the third leg element is movably coupled to the hub **104**. The foot member **106** is coupled to a first end of the first leg element. A second end of the first leg element is movably coupled to a first end of the second leg element. A second end of the second leg element is movably coupled to the first end of the third leg element.

A roller member **108** is coupled to one or more the second end of the second leg element and the first end of the third leg element. The roller element **108** is configured to contact a ground surface when the self-deploying shelter **100** is in the folded or non deployed state.

As is illustrated in FIG. **1**, the assembly **100** is generally in the form of a cross-arched frame structure, with four (4) or more legs **102**. The individual legs **102** are generally comprised of three (3) segments that are configured to be folded in a zigzag pattern. Although the assembly **100** is generally described herein with respect to four legs, the aspects of the disclosed embodiments are not so limited. In alternate embodiments any suitable number of legs can be used other than including four. For example, the assembly **100** can include three or five legs. The number of legs **102** can be dependent upon a variety of factors such as, for example, the size of the assembly **100** and the particular implementation. The legs **102** shown in the example of FIG. **1** can comprise multiple structural elements, also referred to herein as arms. The legs **102** are made up of structural elements, pins, and linkages that move the assembly **100** through a prescribed motion path.

As shown in the example of FIG. **1**, in one embodiment, the assembly **100** includes a hub **104**. The hub **104** will generally include the components and devices that will allow the assembly **100** to transition from an unfolded state to a folded state, and back to the unfolded state. One example of the transition from the folded state to the unfolded state is shown in FIG. **2**.

As is shown in FIG. **2**, the legs **102** are configured to open from the middle and reach their final position before lifting. The legs **102** do not drag on the ground surface during the opening and closing process.

FIG. **2** illustrates five (5) autonomous stages of the deployment of the assembly **100**. In **202**, the assembly **100** is in the folded state. The ends of the legs **102**, also referred to as a foot **106**, are not in contact with the ground surface in the folded state.

In **204**, roller members **108**, more clearly illustrated in FIGS. **3** and **4**, are shown in contact with the ground or floor as the hub **104** causes the legs **102** begin to extend outwards. The roller members **108** are attached to the frame of the assembly **100** using, for example, structural pins to allow for rotation of roller member **108**. In one embodiment, the roller member **108** can comprise a wheel or roller.

As illustrated in the example of FIGS. **3**A and **3**B, examples of the roller member **108** are shown. The roller member **108** can comprise a cam shape or an actual rotating wheel. In alternate embodiments, the roller member **108** can comprise any suitable device that can come into contact with the ground surface as the assembly **100** expands, and provide certain capabilities for movement of the assembly **100** while in contact with the ground surface.

As is also shown in **204** of FIG. **2**, the legs **102** are configured to move outward and upward, away from the ground surface. Once the legs **102** reach a predetermined position during the outward and upward movement, the legs **102** are configured to be moved towards the ground surface until they come in contact with the ground surface.

In one embodiment the assembly **100** includes an integrated four-bar and six-bar linkage system. As the angle between linkages changes through opening, it causes the rest of the assembly, mainly the leg members **102**, to move in relation to the progress of that four-bar and six-bar linkage.

In **206**, the legs **102** are moved towards the ground surface and the feet **106** come in contact with the ground surface. The continued movement of the legs **102** in **208** causes the assembly **100** in this example to lift upwards in the direction of arrow X1 and further away from the ground surface.

In **210**, the assembly **100** is in the fully expanded state. The legs **102** are fully extended and the hub **104** is at the final position of its movement cycle.

FIGS. **3**A and **3**B illustrates an example of a leg **102** in a fully closed position or state. For the purposes of the description below, only one leg **102** will be described. Each leg **102** is configured to connect to the hub **104** at two locations. In one embodiment, there can be a rotating structural pin connection at the upper connection point of the leg **102**, and a rotating structural pin connection at the lower connection point of the leg **102**.

As shown in FIG. **3**A, in the folded state of the assembly **100**, the leg **102** is folded inwards, in a direction toward the hub **104**. The rollers **108** can be configured to support the assembly **100** in the folded state and as the assembly **100** transitions from the folded to unfolded state.

FIG. **4** illustrates a more detailed example of a leg **102** in a partially extended state. In this example, the foot **106** is extended away from the hub **104**. The roller **108** is attached to a middle leg element of the frame. The arrow **402** indicates the movement of the portion of the leg **102** to which the roller **108** is attached as the foot **106** comes into contact with the ground surface as the leg **102** continues to extend outward and downward. The force from the actuation cable or linear actuator causes motion in the four-bar and six-bar linkage which raises the entire assembly.

FIG. **5** illustrates the path **502** of the foot **106** of leg **102** as it moves away from the hub **104** towards the unfolded state. As is illustrated in the example of FIG. **5**, the foot **106** of the leg **102** is never below the roller **108** until it extends to its furthermost position as shown in **206** of FIG. **2**. In this manner, the foot **106** does not scrape against the ground surface during the transition of the assembly **100** from the folded state to the unfolded state.

The example of FIG. **4** illustrates the final foot position where the foot **106** comes in contact with the ground. The leg **102** continues to move and as the leg **102** expands, the hub **104** will start moving upwards in a vertical direction.

Once the legs **102** are in the position shown in FIG. **4**, also referred to as a solid outrigger position, in one embodiment, the feet **106** can be staked. Staking the feet **106** can provide a mechanical advantage during the lift process. Additionally, the fabric that will be put over the assembly **100** can act as lateral stabilizer. In one embodiment, the fabric is typically installed on the frame before the installation or transition sequence. The fabric, including the top, liner, and floor, is able to remain over the frame at all times of the packing, transportation, unpacking, and lifting sequence.

Referring to FIGS. **6**A and **6**B, a strut **602**, such as gas strut, FIG. **6**A or linear actuator, FIG. **6**B, can be connected between the different elements of the leg **102** to assist in the lifting process by providing force to the outer leg element. This strut **602** can either be attached to the assembly **100** via an integrated mounting location or through an additional bracket that is fastened to the member.

FIG. **7** illustrates the nested geometry of the assembly **100** in the folded state. This represents the most packed state of the assembly **100** for storage and shipping. As discussed herein, additional legs are connected to each available point in the hub **104** using fasteners to create the full assembly.

FIG. **8** illustrates one embodiment of a possible actuation system using linear actuators or cables **802** that can be connected to and between the different elements of the legs **102**. By either expanding or contracting, it causes the six bar linkage system to change geometry and lift the assembly. For purposes of the description herein, the cables **802** can also be referred to as ropes or rope paths. In one embodiment, the cables **802** can be connected to an actuator device **806** to control the deployment (folding and unfolding) of the assembly **100**. FIG. **8** shows two actuator devices **806**. Generally, there will be one actuator device **806** per leg **102** of the assembly **100**.

In one embodiment there is one or more sensors on each leg that are configured to provide or sense the angles of each leg **102** relative to the hub **104**. Each leg **102** can be synchronized by keeping the angle of each leg **102** within tolerance of the other legs **102** throughout the opening sequence. Although synchronization is preferred, it is not required. The assembly **100** is configured to be flexible enough to move without full synchronization.

Examples of actuator devices or assemblies **806** are shown in FIGS. **9**A and **9**B. In Figure A, the actuator device **906** includes a rope winch **916** and a drive motor **926**. The rope winch **916** is connected to the cables **802** of FIG. **8** and are driven by the drive motor **926** to control the folding and unfolding of the assembly **100**.

FIG. **9**B shows a linear actuator **906** that elongates or shortens to control the folding and unfolding. The actuator devices **806**, **906** in both examples is disposed between the elements inner member and long link of a leg **102**.

In one embodiment, as shown in FIG. **10**, the cables **802** can include integrated spring tensioners **820** or actively controller linear actuators. The string tensioners can be integrated in line with the cable between the winch and the connection point on the inner member.

FIG. **7** illustrates, the assembly **100** is in the fully closed state. The illustrated elements of the assembly **100** include the hub **104**, leg **1210**, leg **1212**, leg **1214** and leg **1216**. The feet **1224** and **1226**, shown in this example, in contact with the ground surface

FIG. **11** illustrates the assembly **100** in the ready to lift state. In this example, the legs **1210**, **1212**, **1214** and **1216** are extended, and the corresponding feet are in contact with ground. The hub **104** is configured to move in an upward direction.

FIG. **12** illustrates the fully open state. The legs **1210**, **1212**, **1214** and **1216** are fully extended and the hub **104** is has reach its final position.

FIG. **13** illustrates a view of the assembly of a single leg incorporating aspects of the disclosed embodiments. In this example, the assembly includes the rope or cable system **110**, an inner element or member **112**, a long link member **114**, a medium link member **116**, a short link member **118**, and middle member **120**, a roller member **108**, an outer member control linkage system **124**, and outer member **122**, and a foot **106**.

FIG. **14** illustrate an example of an outer member control linkage **124** incorporating aspects of the disclosed embodiments. In this example, the outer member control linkage system **124** includes a crank **1602**, crank **1604**, crank puller **1606**, connector **1608**, brace **1610**, brace **1612**, and brace **1614**.

FIG. **15** illustrates one example of a hub and drive system assembly incorporating aspects of the disclosed embodiments. In this example, the hub and drive system assembly includes drive motor **926**, rope winch **916**, motor shaft coupling **936**, cable pulley **946**, and a High Modulus Polyethylene (HMPE) Rope **902**.

FIG. **16** illustrates an example of a roller **108** incorporating aspects of the disclosed embodiments. In this example, the roller **108** is a pivoting roller assembly. In one embodiment, the roller **108** can include a roller pivot **1802**, a hard stop **1804**, a cam surface **1806**, and a spacer support **1808**.

FIG. **17** illustrates an example of an outer member deployment system assembly incorporating aspects of the disclosed embodiments. In this example, the outer member deployment system assembly includes middle member **120**, connector **1608**, brace **1610**, another brace **1612**, which is a mirror of brace **1610**, second brace **1614**, gas spring **1616**, a mounting bracket **1618**, and an outer member **122**.

FIGS. **18**-**24** illustrate one example of the opening of a rapidly deployable shelter assembly incorporating aspects of the disclosed embodiments.

In FIG. **18**, the assembly is shown beginning to open or unfold. The legs start to go outward, away from the center. The roller helps prevent the feet from scraping the ground during the opening.

In FIG. 19, the assembly is halfway open. The feet do not come in contact with the ground until the assembly is in the ready to lift state.

In FIG. 20, the ready to lift state is reached. The feet are now in contact with the ground. The hub will start to move upwards and the legs will extend or straighten.

FIG. 21 is a side view of FIG. 20. In one embodiment, the feet can be staked at this point and will no longer move.

FIG. 22 illustrates how the rollers have lifted away from the ground. In FIG. 23, the assembly is in the halfway open state. In FIG. 24, the assembly is in the fully open state.

Enhancements to the HEXT structure can include, but are not limited to, hardening for protection and integrating self-powering solar cells to achieve autonomy, power for lighting and electronics, and climate control.

FIGS. 25-28 illustrate various exemplary implementations of a rapidly deployable shelter assembly incorporating aspects of the disclosed embodiments.

Features of the fully autonomous self-deploying shelter assembly 100 of the disclosed embodiments include, but are not limited to:

1. A crossed-arch scheme with four (4) or more legs 102 made of three (3) segments folding in a zigzag pattern.
2. Legs that open from the middle and reach their final position before lifting.
3. Legs that do not drag ever.
4. Legs that are in a solid outrigger position (and staked) for lift with mechanical advantages.
5. The fabric over the structure can act as a lateral stabilizer. The main fabric and floor fabric act as primary structural stabilizers for the frame structure during the installation, the use of the shelter and the dismantling of the shelter, and are an integral part of the structural mechanism and structural load bearing. The stabilization of the structural frame by the fabric during installation allows the frame to be lighter in weight and less rigid at the joints-more flexible and deflect from higher stresses. Aspects of the fabric, such as stiffened areas of fabric, or bungee cords embedded in the fabric membrane are configured to help control the frame folding and are part of the mechanisms.
6. The linear actuator on each leg creates four (4) degrees of freedom.
7. A self-righting capability if the system is on its side initially.
8. Self-balancing, accelerometer-integration.
9. Fully integrated with fabric, floor, and a liner.
10. Can be remotely operated or operated in person.
11. The assembly 100 is configured to walk and move itself to an optimal position, find level ground using integrating angle sensors. Each leg 102 has the ability to move independently. Therefore, one leg can close partially at a time, creating a sequence where the system closes partially, leans to one side, and extends out for movement.
12. Multiple units can walk to each other to automatically connect and sync up.
13. Has a modular actuation system for one (1) or multiple degrees of freedom.
14. Has a minimal frame and minimum weight configuration.
15. Is made of folded aluminium panels or extruded metals or composites. The folded legs nest within one another with structural stability.
16. Has a lightweight and efficient structural design.
17. Has a unique ornamental design.
18. Has a six (6)-bar linkage system combined with a seven (7)-bar linkage system.
19. Includes control logic for opening and closing, error control and system integration.
20. Is fully self-powered with self-powering solar integration.
21. Can be used with rigid flooring. Also has an integrated folded floor concept.
22. There is control of the fabric folding when closing by using embedded elastic members.
23. There is integration of Teflon washers and unique pin joints.
24. Uses cable actuation to control system deployment.
25. The integrated pull-bar springs support stability and allow continuous opening for self-balancing.
26. Adaptable for a larger-scale truck/trailer mounted hydraulic system.
27. Also adaptable for airdrop deployment, autonomous vehicle dropping and parachute deployment.
28. Enables the auto-build of forward operating base(s) or deployable tent cities without human intervention.
29. The system self-closes.
30. Links are connected using driftpin shoulder bolt attachments.
31. Passive gas spring stabilization is used during opening and closing of the legs.
32. The system has amobile power unit and control box with weatherproof receptacles.
33. The system includes a stop-start quick disconnect controller.
34 The system includes an E-stop (emergency stop) button for safety during building and deployment.

While there have been shown, described and pointed out, fundamental novel features of the invention as applied to the exemplary embodiments thereof, it will be understood that various omissions, substitutions and changes in the form and details of devices and methods illustrated, and in their operation, may be made by those skilled in the art without departing from the spirit and scope of the presently disclosed invention. Further, it is expressly intended that all combinations of those elements, which perform substantially the same function in substantially the same way to achieve the same results, are within the scope of the invention. Moreover, it should be recognized that structures and/or elements shown and/or described in connection with any disclosed form or embodiment of the invention may be incorporated in any other disclosed or described or suggested form or embodiment as a general matter of design choice.

What is claimed is:

1. A self-deploying shelter assembly configured to deploy from a folded state to an unfolded state and from the unfolded state to the folded state, the shelter assembly comprising:

a plurality of legs;

a hub, the hub being configured to be coupled to one end of respective legs of the plurality of legs, wherein a leg of the plurality of legs comprises:

a first leg element, a second leg element and a third leg element, a second end of the third leg element being movably coupled to the hub;

a foot member coupled to a first end of the first leg element, a second end of the first leg element being movably coupled to a first end of the second leg element and a second end of the second leg element being movably coupled to the first end of the third leg element; and a roller member coupled to one or more of the second end of the second leg element and the first end of the third leg element, the roller member being configured to contact a ground surface when the self-deploying shelter is in the folded state; and

when the hub member causes the plurality of legs to extend outwards, the first leg element and the second leg element extend outwards a pre-determined distance until the foot member comes in contact with a ground surface.

2. The shelter assembly according to claim **1**, wherein the hub is configured to:

cause a movement of the plurality of legs outward, away from the hub, when the shelter assembly is deploying from the folded state to the unfolded state; and

cause a movement of the plurality of legs inward, toward the hub, when the shelter assembly is deploying from the unfolded state to the folded state.

3. The shelter assembly according to claim **1**, wherein the roller member comprises a curved member that extends from a bottom of the shelter assembly to make contact with the ground surface when the shelter assembly is in the folded state.

4. The shelter assembly according to claim **1**, wherein the foot member is away from the ground surface in the folded state of the shelter assembly.

5. The shelter assembly according to claim **1**, wherein the contact of the foot member with the ground surface while the hub is extending the plurality of legs causes the hub to move in an upwards direction, away from the ground surface.

6. The shelter assembly according to claim **1**, wherein the foot member is disposed closer to the hub than the roller member in the folded state of the assembly.

7. The shelter assembly according to claim **1**, wherein the first leg element is pivotally connected to the foot member and the second leg element.

8. The shelter assembly according to claim **1**, wherein the second leg element is pivotally connected to the third leg element.

9. The shelter assembly according to claim **1**, wherein the roller member comprises a bracket that is pivotally connected to the third leg element and the second leg element.

10. The shelter assembly according to claim **1**, wherein the foot member moves along an arced path when moving to and from the folded state and unfolded state of the shelter assembly.

11. The shelter assembly according to claim **10**, wherein when the shelter assembly deploys from the folded state to the unfolded state, the hub is configured to cause:

the first end of the first leg element to move in an upward direction away from the ground surface, wherein the first end of the first leg element is moving along the arced path;

the first end of the second leg element to move away from a centerline of the hub and third leg element;

the foot member to come in contact with the ground surface when the first leg element and the second leg element reach pre-determined positions;

the roller member to move away from the ground surface after the foot member comes in contact with the ground surface and the third leg element extends;

the shelter assembly to reach the unfolded state once the third leg element is fully extended.

12. The shelter assembly according to claim **10**, wherein when the shelter assembly deploys from the unfolded state to the folded state, the hub is configured to cause:

the third leg element to retract from a fully extended position until the roller member comes in contact with the ground surface;

the foot member to lift off the ground surface after the roller member comes in contact with the ground surface; and

the second leg element and the first leg element to retract to the folded state of the shelter assembly, while the foot member moves along the arched path, without further contact with the ground surface.

13. The shelter assembly according to claim **1**, wherein the third leg element comprises a first leg member spaced apart from a second leg member, a first end of the first leg member and a first end of the second leg member are pivotally connected to the hub, and a second end of the first leg member is coupled to a roller arm bracket.

14. The shelter assembly according to claim **13** further comprising an actuator or cable assembly connected to and between the hub, the first leg member and the second leg member, the actuator or cable assembly being configured to move the first leg member and the second leg member when extended and retracted by a spool assembly on the hub.

15. The shelter assembly according to claim **14** further comprising a drive motor connected to the hub and the spool assembly, the drive motor configured to cause the spool assembly to extend and retract a cable member.

16. The shelter assembly according to claim **1**, further comprising a linkage system coupled to the plurality of legs, the linkage system configured to cause an individual leg to open to the unfolded state and close to the folded state along a prescribed path.

17. The shelter assembly according to claim **16**, wherein the linkage system is connected between the hub and the individual leg of the plurality of legs.

18. The shelter assembly according to claim **16**, wherein the linkage system is connected to an actuator or cable assembly.

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